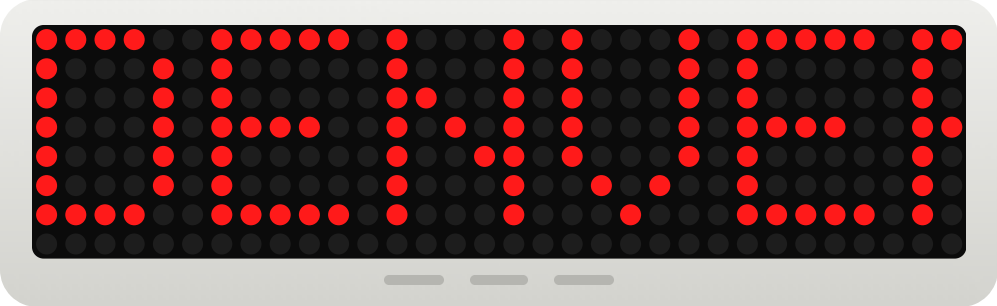
QUICK START GUIDE

# Overhead

A pixel clock that shows where the plane over your house came from, or where it's going, only while you can actually hear it.



A Delta flight from Denver passing overhead. Between planes it's a clock.

## What you need

**Ulanzi TC001**
The pixel clock (about $40–50). Nothing to open or solder.

**Your phone**
To put the clock on your Wi-Fi and to use its settings page.

**A Windows PC**
For the one-click installer. Or any browser, using the clock's update page.

## Three steps, all over Wi-Fi

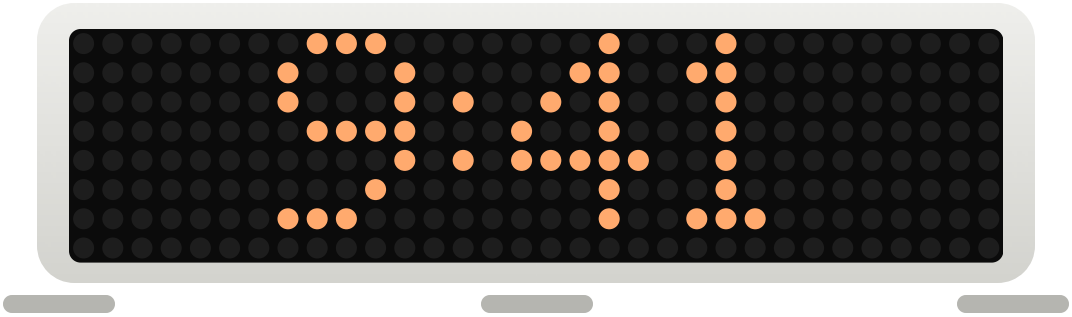
1. **Put the clock on your Wi-Fi**
   Plug it in. A new TC001 shows `192.168.4.1`: join the clock's Wi-Fi network on your phone, open `http://192.168.4.1` and choose your home network. The clock then joins it.

2. **Install Overhead**
   On a Windows PC on the same Wi-Fi, download `overhead-installer.zip` from the project's Releases page on GitHub (github.com/MishaTikh/overhead-clock), unzip it and double-click **Install-Overhead.bat**. It finds the clock and installs Overhead in about 30 seconds. On a Mac, Linux or a phone: open `http://clock.local/update`, choose `overhead-tc001-app.bin` and upload.

3. **Sign in and set your address**
   Hold the clock's middle button for 3 seconds, until the display changes: it scrolls its settings address, then the 12-character password four characters at a time (`K7PR`, `3XQ9`, `TMWA`), each standing still. The left and right buttons step back and forward. Open that address, sign in as `admin` and type your street address.

USING IT

## Three buttons

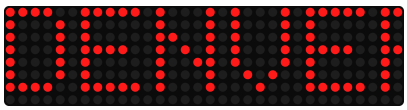
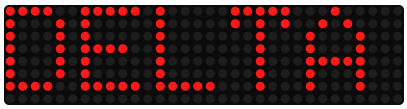
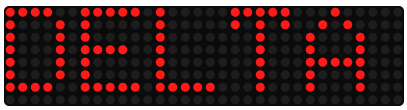


**Left**
Press: what to show for each plane (cycles the four styles below).
Hold: replay recent planes.
Held while plugging in: forget Wi-Fi.

**Middle**
Press: switch between *Clock + planes* and *Planes only*.
Hold: show the settings address and password.

**Right**
Press: brightness, cycling Auto, Low, Medium, High.
Auto follows the room light.

## What it shows for each plane

| | | |
|---|---|---|
| **City** | Where a landing plane came from, or where a departing plane is going. The default. |  |
| **Codes** | Airport codes, from and to. |  |
| **Airline** | Airline name with the codes. |  |
| **Full** | Everything: flight, cities and aircraft type. |  |

Long text scrolls. If a flight's route isn't known, every style shows the flight number. Colors follow the airline (Delta red, United blue, Sun Country orange…).

## Two modes

**Clock + planes** (default)
Shows the time; switches to the flight while a plane is audible, then back.

**Planes only**
Stays dark until a plane is audible. Nice for a bedroom or a shelf you don't want glowing.

SETTINGS PAGE

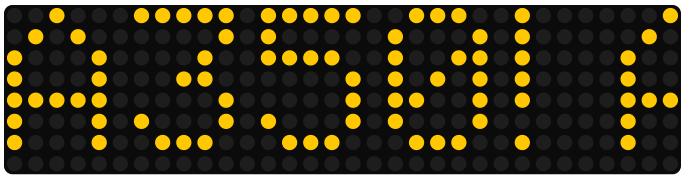
## Everything else is on your phone

Open `http://overhead.local` (user `admin`). Changes save as you make them. Some phones don't open `.local` names: hold the middle button to see the clock's number instead.

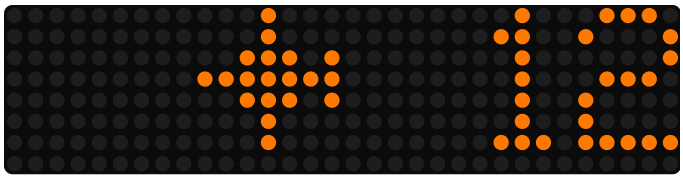
| SECTION | WHAT YOU CAN DO |
|---|---|
| **Today** | Planes heard today, the farthest origin, the loudest plane, rare visitors, and an hour-by-hour chart against a usual day (learned over the first week). |
| **Airport weather** | Temperature and wind at your airport, and whether the wind has planes *landing over you*, *taking off over you* or going elsewhere today. |
| **Home location** | Street address (looked up once) or coordinates if you'd rather not send an address anywhere. Airport (nearest by default) and time zone (automatic). |
| **What the sign shows** | Mode, display style, 12 or 24-hour clock. |
| **Alerts** | Rare aircraft in gold with a sparkle; an optional chime; an inbound countdown with a little plane icon; an evening recap at the hour you choose. |
| **Clock screen extras** | Every so often, briefly show: indoor temperature and humidity, outdoor weather, date, sunrise and sunset, whether planes are over you today, planes heard today. All off by default. |
| **Colors** | Clock color, airline colors on or off, and a color for other planes. |
| **Brightness and text** | Auto, low, medium or high; dark-room and daylight levels; scroll speed; try any text on the sign. |
| **Quiet hours** | No plane alerts or data requests overnight (11 pm–6 am by default). The clock keeps showing the time. |
| **Sensitivity** | How loud a plane must be to show (60 dBA is clearly noticeable over a quiet street); daily data budget; saving requests when planes are elsewhere. |
| **Wi-Fi and security** | Network and signal, change the admin password, forget Wi-Fi. |

### Getting the sensitivity right

For the first few days, compare the *Recent* list with what you hear. If the sign lights before a plane is clearly audible, raise the threshold a few dBA; if it misses planes you notice, lower it.



A rare visitor: gold, with a sparkle.



Inbound countdown: a plane is 12 seconds from audible.

GOOD TO KNOW

## Troubleshooting

| IF… | TRY THIS |
|---|---|
| The installer can't find the clock | Make sure the PC and clock are on the same Wi-Fi. Type the clock's address when the installer asks (hold the middle button to see it). |
| Updating later | Run the installer again, or use *Update firmware* on the settings page. |
| Your phone doesn't open the setup page | Stay on the clock's Wi-Fi and browse to `192.168.4.1`. |
| You changed Wi-Fi or moved | Hold the left button while plugging the clock in. It forgets the old network and shows the setup Wi-Fi again. |
| `overhead.local` doesn't open | Hold the middle button; open the number it scrolls, like `192.168.1.42`. |
| Forgot the admin password | Hold the middle button: it shows the password four characters at a time; left and right step through it. |
| It shows flight numbers, not cities | The route for that flight isn't known yet. Common at airports without a supported flight board; add one for yours (see CONTRIBUTING on GitHub). |
| Nothing ever shows | Check the settings page: address found? Airport nearby? Inside quiet hours? Planes may be using other runways today (see Airport weather). |
| Wi-Fi or the internet is down | Nothing to do: it keeps working as a clock and picks up planes again when the connection returns. |

## Privacy and security

**Stays at home**
No account, cloud or remote access, and no internet port opened. The setup Wi-Fi, the settings page and firmware updates are all password protected.

**What leaves the house**
Your address once (or nothing, if you enter coordinates), your location rounded to about 1 km when asking for nearby planes, and flight numbers when looking up routes. Every connection is verified HTTPS.

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